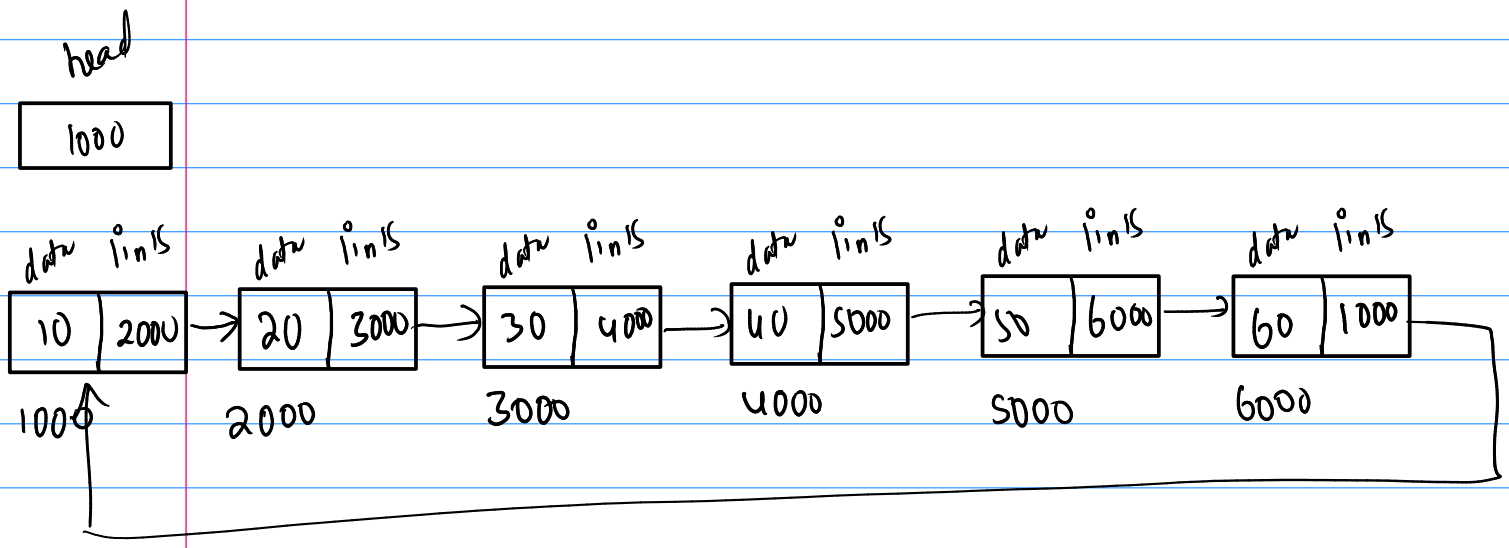
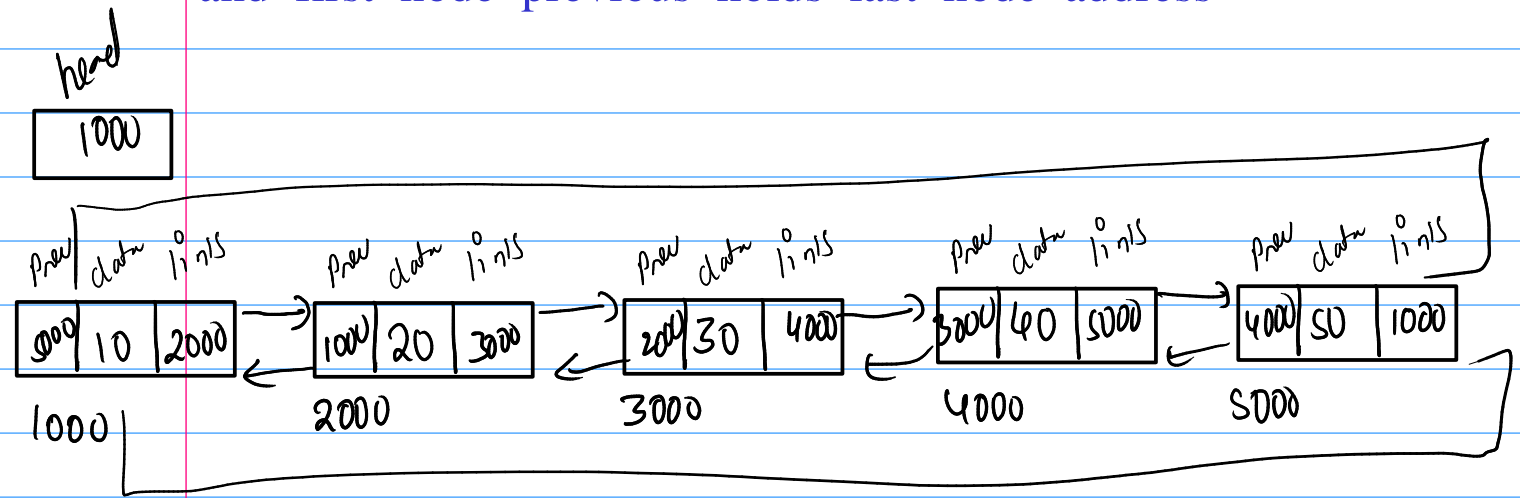
3/04 Circular linked list

circular singly link list
circular doubly link list

In Circular singly link list last node link will hold first node address



In circular doubly link list last node lin holds first node address and first node previous holds last node address



Delete based on given data

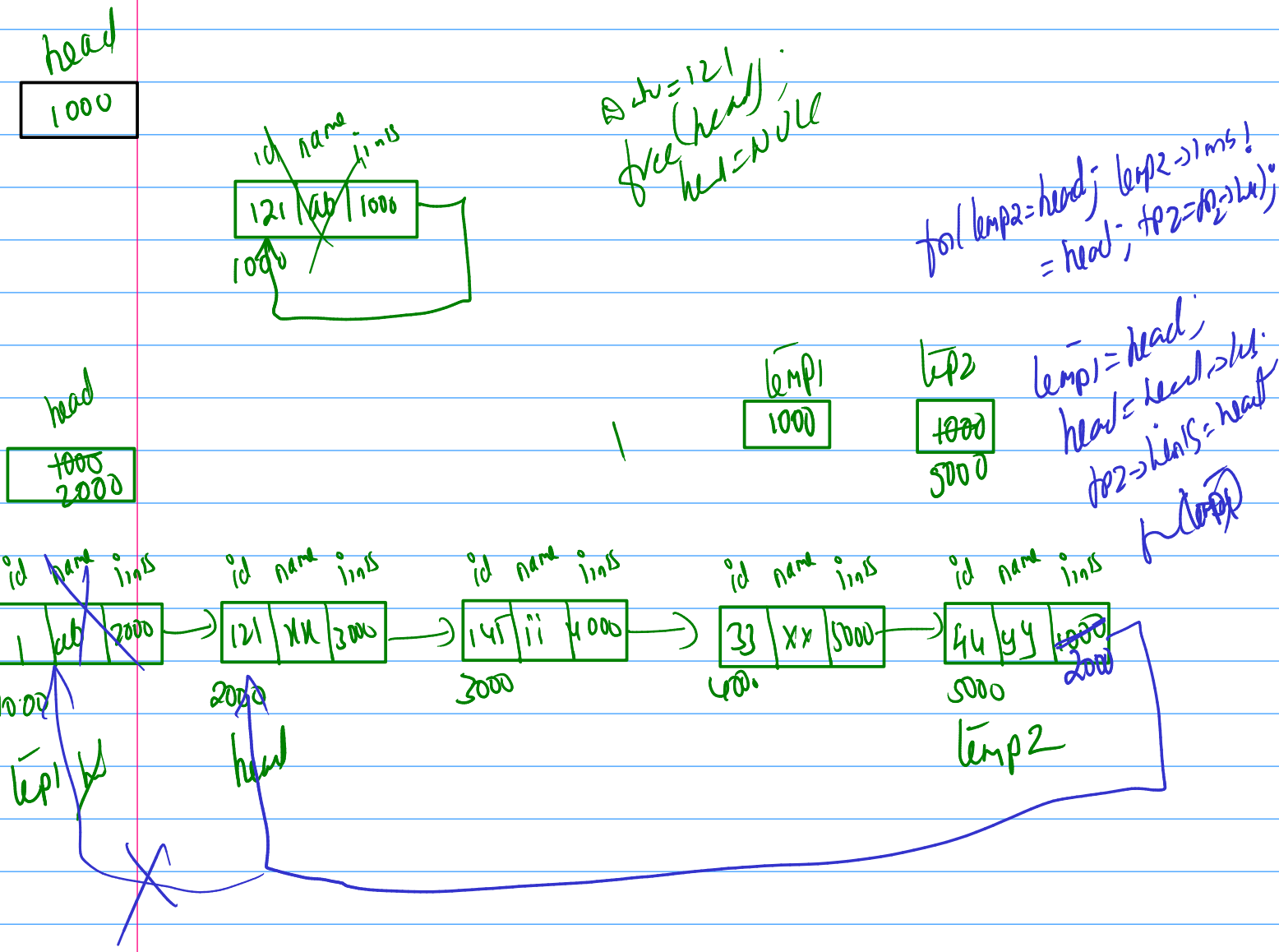
1. list is emtpy , print list empty msg
2. else if data matching with first
   a. list having only one node, delete the head and assign NULL to head
   b. else traverse up to last node

assign head to temp, move head to next node, assign head to la node link, then free temp

3. else traverse until we reach the first node or until data match
4. Once loop termainted check we reached first node

data not found

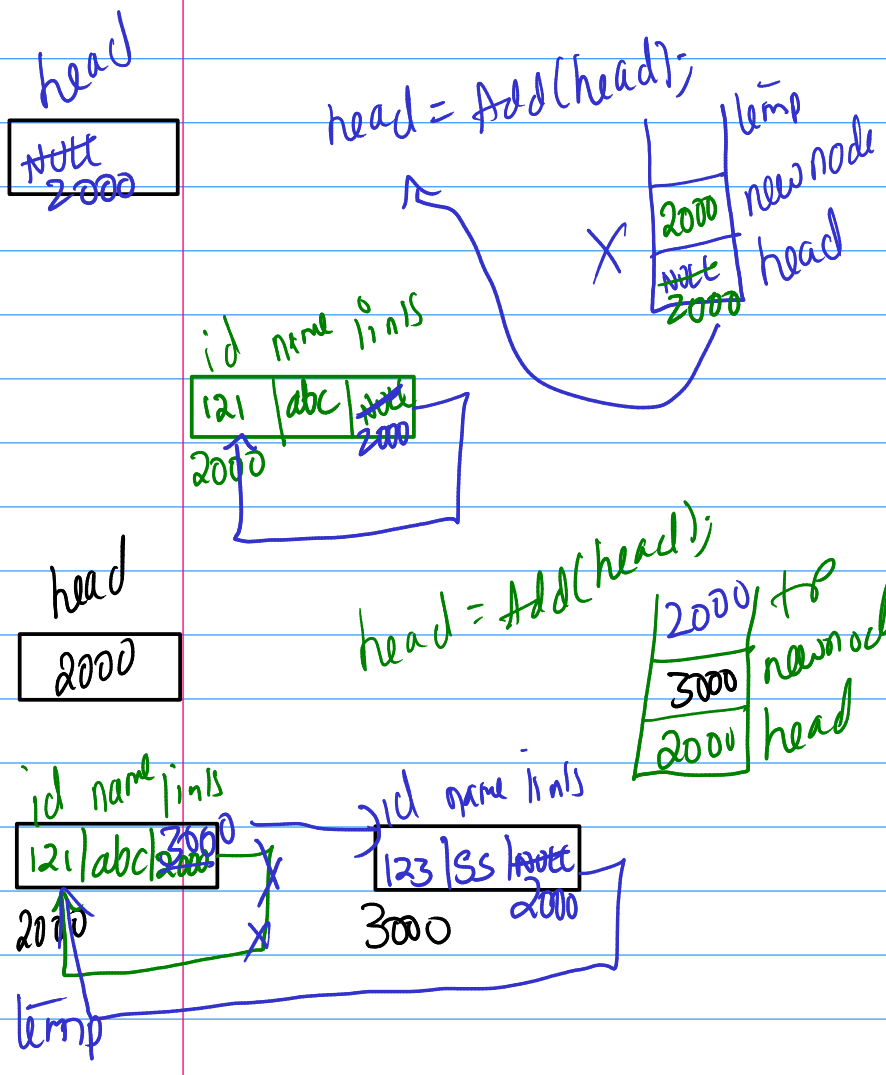
5. else link previous node and next node of node to deleted
6. return first node address

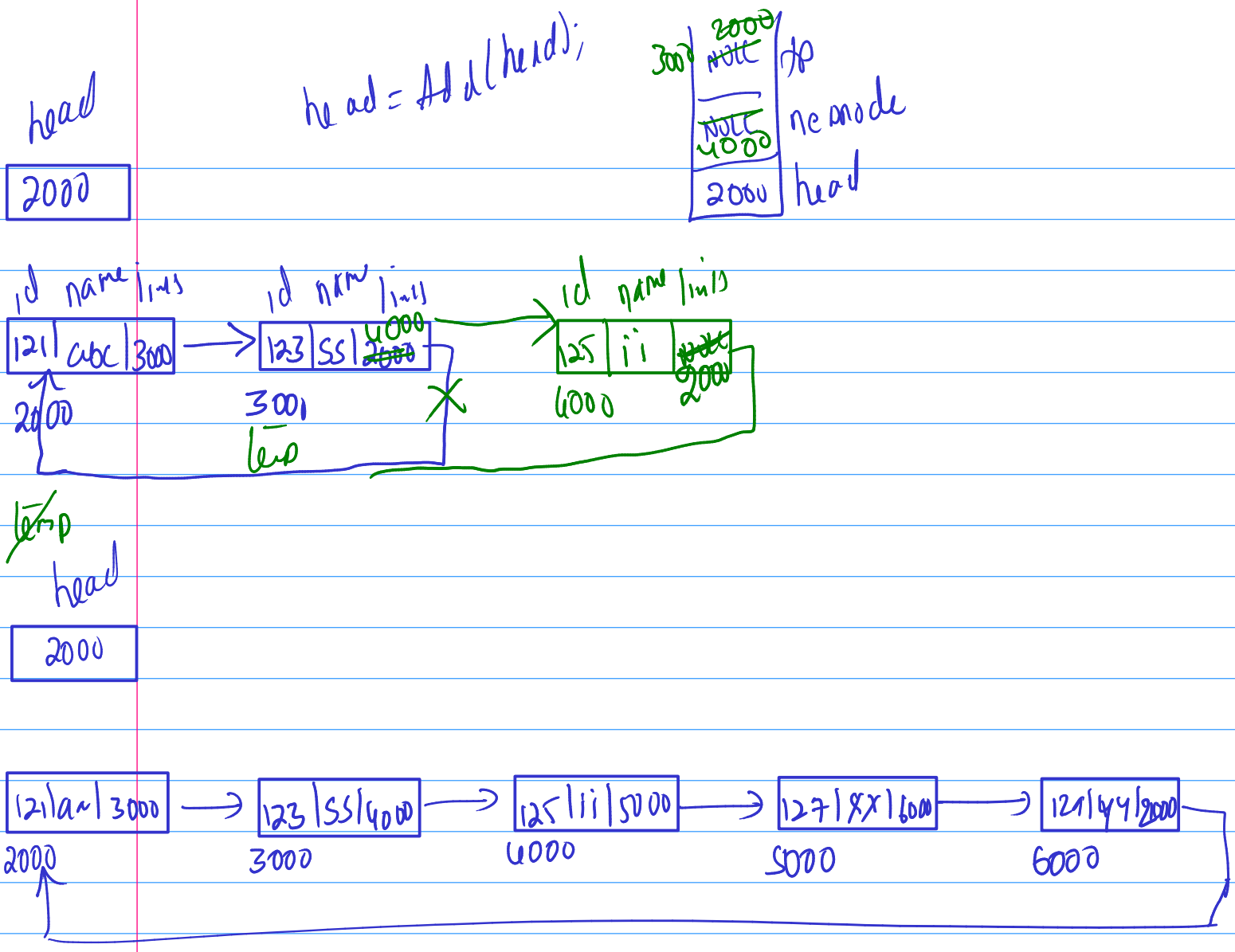


Circular Singly Linked List

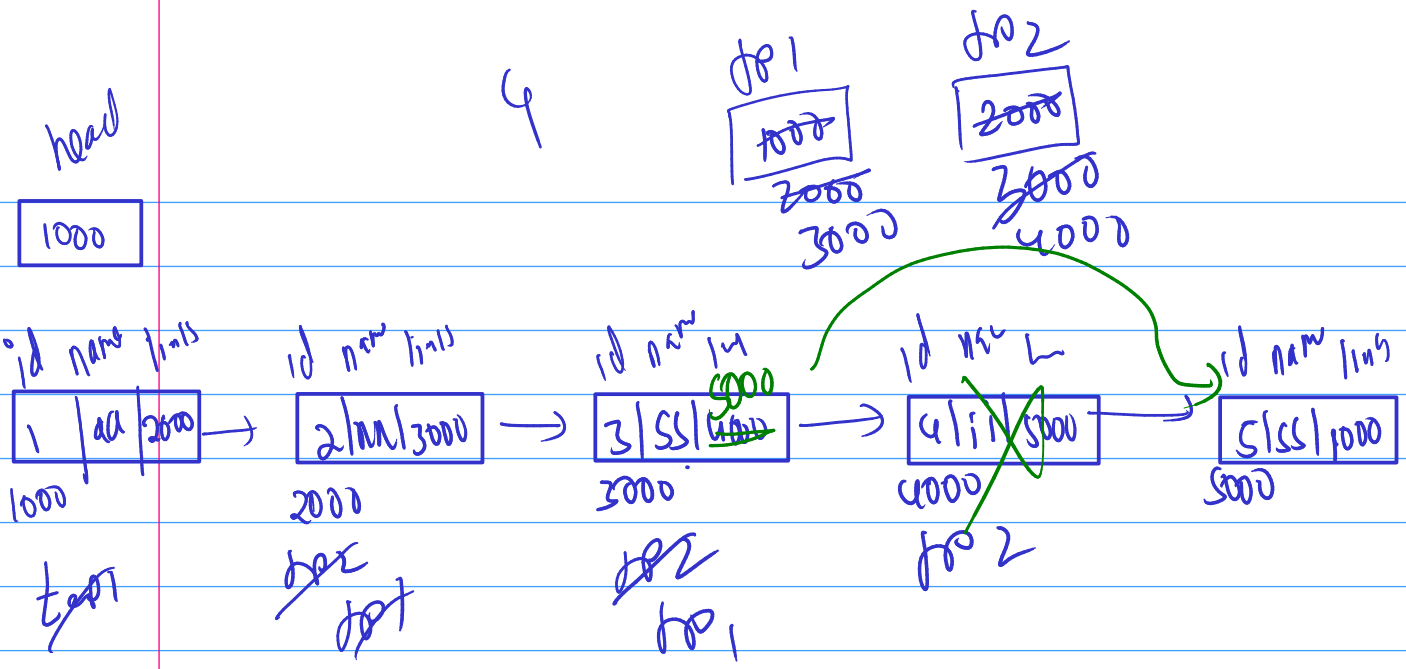
Algorithm for AddatLast

1. creation of node
2. initialise the node
3. link to list
   if list is empty, assign newnode to head
   else traverse upto last node , assign newnode to last node link
4. assign head to newnode link
5. return first node address





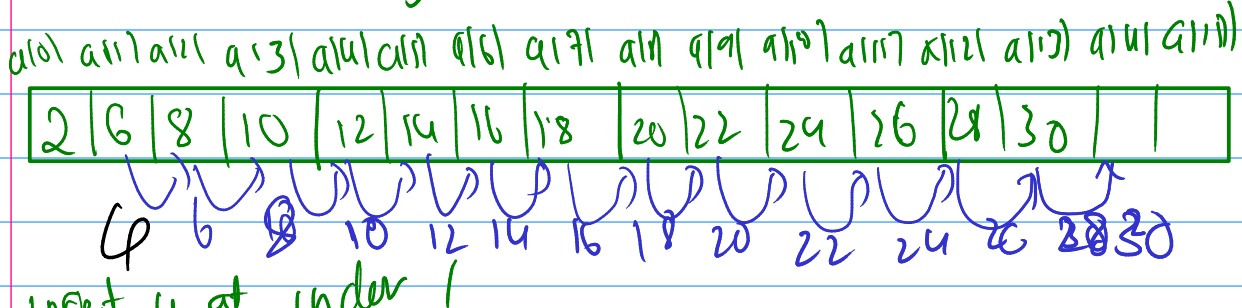

```
temp = head;
do
{ pf("%d %s\n", temp->id, temp->name);
  temp = temp->link;
} while (temp != head);
```



Advantages of Arrays

Single name to represent group of values.
Random accessing of an element is O(1)
adding element at last index takes O(1)
deleting last element of an array takes O(1)

Drawbacks :
fixed size
random insertion
random deletions
} becomes costlier process



Demands contiguous locations